Camera Dissection: Reverse Engineering in Action

MAE1001

Fall 2025

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Course webpage: https://gwu-mae1001.github.io/fall2025/

Content

Learning Objectives

Students will get **"Shocked"** and learn:

Reverse Engineering

Learn basics of reverse engineering through hands-on dissection.

2. Core Components

Understand core electronic components inside a disposable camera.

3. Ethical AI

Apply ethical AI to identify parts. Use AI as starting point, not as your answer.

History

Timeline

The evolution of cameras reflect innovation, cost, and speed to improve accessibility & engineering

1816

Camera Obsucra

Joseph Nicéphore Niépce captures the first permanent image using a camera obscura (a "dark room" box with a pinhole lens). Light projected through the hole etched an image onto light-sensitive material. This marks the foundation of photography.

1888

Film Camera

George Eastman's Kodak camera introduces roll photographic film. Instead of glass plates, light passed through a lens onto film coated in silver-halide crystals, which chemically reacted to create an image. Made photography portable and affordable for the public.

1948

Disposable Camera

Alfred D. Weir designs the first single-use camera in Dallas. Made of inexpensive fiberboard and pre-loaded with 35mm film. A simple shutter and plastic lens exposed film, while a small capacitor-powered flash enabled indoor shots. It was cheap, compact, and easy to buy anywhere.

1975

Digital Camera

Steven Sasson at Kodak builds the first digital camera prototype using a CCD (charge-coupled device) sensor. Instead of film, light hitting the sensor was converted into electrical signals and stored digitally on a cassette tape. This invention laid the groundwork for modern digital imaging.

Why this lab

Annual Tradition @ GWU

Reverse Engineering Lab

- Annual intro labs for new students
- Learn simple engineering concepts
- Experience Reverse Engineering
- Learn documentation skills

What’s New

- Use of Ethical AI to identify each components to blend tradition with new technology
- Introduce modern tools and ethics

GWU Tradition

- Experience the capacitor “pop” and shock fellow students

CAUTION STEPS

How it Works

BE CAREFUL - it hurts :)

- Don't Touch Any Metal parts

  Hold the Battery Side

  Avoid The 2 Silver Prongs on Capacitor
- Create Spark with Screwdriver
- Once Sparked it has no charge
  - Recharge by placing Battery back and pressing the copper lever in back

https://drive.google.com/file/d/1ZB5xcZ0gM5Sy0ifMQdOAkQVqb5P1VthZ/view?usp=sharing

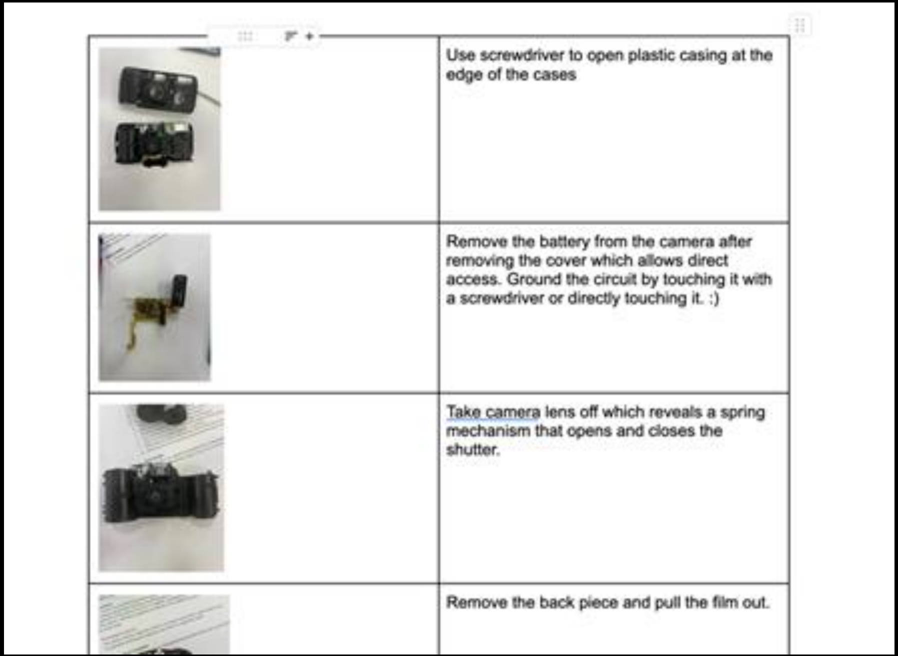
Hold Here

What is a Lab Notebook

Document each step by writing what you did and adding images to help understand.

Steps:

1. Create a Google Doc for the team
2. Write down each step that you take and your thoughts
3. Take a lot of pictures during the process with your phone
4. Share it with the team
5. Submit



	Use screwdriver to open plastic casing at the edge of the cases
	Remove the battery from the camera after removing the cover which allows direct access. Ground the circuit by touching it with a screwdriver or directly touching it. :)
	Take camera lens off which reveals a spring mechanism that opens and closes the shutter.
	Remove the back piece and pull the film out.

How the lab will work

**Open camera and take a picture! (Document steps and add picture of each step) Create Google Doc

1

Remove wrapper and stickers on camera

Reveal all the hinges and where the camera can be opened to reveal internal sections.

2

Remove battery

Ensure a couple pictures were taken to activate the capacitor, but remove battery to not overcharge Swing open the three sections on the bottom.

3

Dissect

Use a screwdriver to force open the camera at the opening lines (Might be difficult). Take the electrical component out while watching where you touch

4

Shock

Record a video of this happening!

Discharge the capacitor using a screwdriver and watch it pop. (Touch it but it hurts..)

5

Dissect

Remove remaining components found in the camera. Start from the back cover, then the clear plastic on top, gears, springs, etc...

Submit Videos Of You Getting "Shocked"

https://forms.gle/LiaAWtt9V2NctS2s6

Lab Notebook

Deliverable 1

- Record each step and observation including photos
- Include each identified component

Ethical AI

Deliverable 2

Include in Lab Notebook

- Take photo of every component lined up
- Input photo into AI tool and create a prompt to help identify each component
- Paste AI prompt and responses into exit ticket
- Delve deeper and ask questions into how it works

Thank you

Ready for what's next?

Come to Office Hours

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